

Problem Set 5—Suggested Answer

Exercises

1. Identification failure with a single instrument.

Substitute the two reduced forms $X_{i1} = \pi Z_i + v_i$ and $X_{i2} = \gamma Z_i + w_i$ into the structural equation:

$$\begin{aligned} Y_i &= \beta_1(\pi Z_i + v_i) + \beta_2(\gamma Z_i + w_i) + e_i \\ &= (\beta_1\pi + \beta_2\gamma) Z_i + \underbrace{(e_i + \beta_1 v_i + \beta_2 w_i)}_{\eta_i}. \end{aligned}$$

The composite error η_i is uncorrelated with Z_i because

$$E(Z_i \eta_i) = \underbrace{E(Z_i e_i)}_{=0} + \beta_1 \underbrace{E(Z_i v_i)}_{=0} + \beta_2 \underbrace{E(Z_i w_i)}_{=0} = 0.$$

Here $E(Z_i e_i) = 0$ is the instrument exogeneity condition, while $E(Z_i v_i) = 0$ and $E(Z_i w_i) = 0$ hold because v_i and w_i are population regression residuals.

Therefore a regression of Y_i on Z_i consistently estimates the single coefficient $\delta := \beta_1\pi + \beta_2\gamma$. Even if we have consistent estimates of π and γ from the two reduced forms, we still face *one equation with two unknowns* (β_1, β_2) and cannot separately identify them.

Contrast with two instruments. With two instruments, OLS on the reduced forms (1) gives consistent estimates of $\pi_1, \pi_2, \gamma_1, \gamma_2$, and OLS on the structural reduced form (2) gives consistent estimates of $\delta_1 := \beta_1\pi_1 + \beta_2\gamma_1$ and $\delta_2 := \beta_1\pi_2 + \beta_2\gamma_2$. This yields the linear system

$$\begin{pmatrix} \delta_1 \\ \delta_2 \end{pmatrix} = \begin{pmatrix} \pi_1 & \gamma_1 \\ \pi_2 & \gamma_2 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix},$$

which is *two equations in two unknowns*. Provided the rank condition holds (the 2×2 matrix above is invertible), we can uniquely solve for β_1 and β_2 . This illustrates why the number of instruments must be at least as large as the number of endogenous regressors.

2. Upper triangular block structure of π .

Let $K_1 = \dim(X_{i1})$ and $L_2 = \dim(Z_{i2})$. Partition conformably with $Z_i = (X'_{i1}, Z'_{i2})'$ and $X_i = (X'_{i1}, X'_{i2})'$:

$$A := E(Z_i Z_i') = \begin{pmatrix} E(X_{i1} X_{i1}') & E(X_{i1} Z_{i2}') \\ E(Z_{i2} X_{i1}') & E(Z_{i2} Z_{i2}') \end{pmatrix},$$

$$B := E(Z_i X_i') = \begin{pmatrix} E(X_{i1} X_{i1}') & E(X_{i1} X_{i2}') \\ E(Z_{i2} X_{i1}') & E(Z_{i2} X_{i2}') \end{pmatrix},$$

and partition π in the same way:

$$\pi = \begin{pmatrix} \pi_{11} & \pi_{12} \\ \pi_{21} & \pi_{22} \end{pmatrix}.$$

The claim is $\pi_{21} = 0_{L_2 \times K_1}$. From the hint, it suffices to show that $\pi_{11} = I_{K_1}$ and $\pi_{21} = 0$ satisfies $A\pi = B$. Isolate the *first block-column* of $A\pi = B$:

$$\begin{pmatrix} E(X_{i1}X'_{i1}) & E(X_{i1}Z'_{i2}) \\ E(Z_{i2}X'_{i1}) & E(Z_{i2}Z'_{i2}) \end{pmatrix} \begin{pmatrix} \pi_{11} \\ \pi_{21} \end{pmatrix} = \begin{pmatrix} E(X_{i1}X'_{i1}) \\ E(Z_{i2}X'_{i1}) \end{pmatrix}.$$

Substitute $\pi_{11} = I_{K_1}$ and $\pi_{21} = 0$:

$$\text{Row 1: } E(X_{i1}X'_{i1}) \cdot I_{K_1} + E(X_{i1}Z'_{i2}) \cdot 0 = E(X_{i1}X'_{i1}) \quad \checkmark$$

$$\text{Row 2: } E(Z_{i2}X'_{i1}) \cdot I_{K_1} + E(Z_{i2}Z'_{i2}) \cdot 0 = E(Z_{i2}X'_{i1}) \quad \checkmark$$

Both rows are satisfied. Since A is invertible (rank condition), the solution to $A\pi = B$ is unique, so we must have $\pi_{11} = I_{K_1}$ and $\pi_{21} = 0_{L_2 \times K_1}$, giving the upper triangular block structure

$$\pi = \begin{pmatrix} I_{K_1} & \pi_{12} \\ 0 & \pi_{22} \end{pmatrix}.$$

Interpretation. The bottom-left block $\pi_{21} = 0$ says that in the population regression of X_{i1} on $Z_i = (X'_{i1}, Z'_{i2})'$, the external instruments Z_{i2} receive zero weight. This is intuitive: X_{i1} is already included in Z_i , so it is its own best predictor; the external instruments Z_{i2} add nothing for explaining X_{i1} .

3. Attenuation bias from measurement error.

We observe $\tilde{X}_i = X_i + r_i$ but the true regressor is X_i . The OLS estimator from regressing Y_i on $\tilde{X}_i$ is (using $E(X_i) = E(r_i) = 0$ so no demeaning is needed):

$$\hat{\beta}^{\text{OLS}} = \frac{\sum_{i=1}^N \tilde{X}_i Y_i}{\sum_{i=1}^N \tilde{X}_i^2}.$$

Rewrite Y_i in terms of $\tilde{X}_i$:

$$Y_i = \beta X_i + e_i = \beta(\tilde{X}_i - r_i) + e_i = \beta \tilde{X}_i + (e_i - \beta r_i).$$

Substituting:

$$\hat{\beta}^{\text{OLS}} = \frac{\sum_{i=1}^N \tilde{X}_i [\beta \tilde{X}_i + (e_i - \beta r_i)]}{\sum_{i=1}^N \tilde{X}_i^2} = \beta + \frac{\sum_{i=1}^N \tilde{X}_i (e_i - \beta r_i)}{\sum_{i=1}^N \tilde{X}_i^2}.$$

Divide numerator and denominator by N and apply the WLLN:

$$\frac{1}{N} \sum_{i=1}^N \tilde{X}_i^2 = E[(X_i + r_i)^2] + o_p(1) = \sigma_X^2 + \underbrace{2E(X_i r_i)}_{=0} + \sigma_r^2 + o_p(1) = \sigma_X^2 + \sigma_r^2 + o_p(1),$$

Exercise 2

multivariate regression

$$X_i = \pi' z_i + v_i$$

in blocks:

$$\begin{aligned} \begin{pmatrix} X_{1i} \\ X_{2i} \end{pmatrix} &= \begin{pmatrix} \pi_{11} & \pi_{21} \\ \pi_{12} & \pi_{22} \end{pmatrix} \begin{pmatrix} X_{1i} \\ z_{2i} \end{pmatrix} + v_i \\ &= \begin{pmatrix} \pi_{11} X_{1i} + \pi_{21} z_{2i} \\ \pi_{12} X_{1i} + \pi_{22} z_{2i} \end{pmatrix} + v_i \end{aligned}$$

suggests that $\pi_{11} = I$ and $\pi_{21} = 0$

where $E(X_i r_i) = E(X_i)E(r_i) = 0$ by independence and zero means.

For the numerator, expand $\tilde{X}_i = X_i + r_i$:

$$E[\tilde{X}_i(e_i - \beta r_i)] = \underbrace{E(X_i e_i)}_{=0} + \underbrace{E(r_i e_i)}_{=0} - \beta \underbrace{E(X_i r_i)}_{=0} - \beta E(r_i^2) = -\beta \sigma_r^2.$$

Here $E(X_i e_i) = 0$ follows from $E(e_i|X_i) = 0$ via the law of iterated expectations, and the terms involving r_i paired with e_i or X_i vanish by the independence of r_i from (X_i, e_i) .

Combining, and using $o_p(1)$ arithmetic throughout:

$$\hat{\beta}^{\text{OLS}} = \beta + \frac{-\beta \sigma_r^2 + o_p(1)}{\sigma_X^2 + \sigma_r^2 + o_p(1)} = \beta \left(1 - \frac{\sigma_r^2}{\sigma_X^2 + \sigma_r^2} \right) + o_p(1).$$

Therefore

$$\text{plim } \hat{\beta}^{\text{OLS}} = \beta \cdot \frac{\sigma_X^2}{\sigma_X^2 + \sigma_r^2}.$$

Special cases.

- $\sigma_r^2 = 0$: $\text{plim } \hat{\beta}^{\text{OLS}} = \beta$. No measurement error means $\tilde{X}_i = X_i$, so OLS is consistent.
- $\sigma_r^2 \rightarrow \infty$: $\text{plim } \hat{\beta}^{\text{OLS}} \rightarrow 0$. The measurement error swamps the signal in X_i ; the estimate is pulled to zero regardless of β .

The ratio $\sigma_X^2 / (\sigma_X^2 + \sigma_r^2) \in (0, 1)$ is the *signal-to-total-variance* ratio. It acts as a shrinkage factor that always attenuates the estimate toward zero, hence the name *attenuation bias* (more precisely, *attenuation inconsistency*: we are studying the probability limit, not the expectation).

4. Consistency of $\hat{\beta}^{\text{IV}}$.

Setup: $Y_i = X_i' \beta + e_i$ with $E(e_i Z_i) = 0$ and $E(Z_i X_i')$ invertible (rank condition, just-identified case). The IV estimator is

$$\hat{\beta}^{\text{IV}} := \left(\frac{1}{N} \sum_{i=1}^N Z_i X_i' \right)^{-1} \frac{1}{N} \sum_{i=1}^N Z_i Y_i.$$

Step 1: Decompose $\hat{\beta}^{\text{IV}} - \beta$. Substitute $Y_i = X_i' \beta + e_i$:

$$\frac{1}{N} \sum_{i=1}^N Z_i Y_i = \frac{1}{N} \sum_{i=1}^N Z_i X_i' \beta + \frac{1}{N} \sum_{i=1}^N Z_i e_i,$$

so

$$\hat{\beta}^{\text{IV}} - \beta = \left(\frac{1}{N} \sum_{i=1}^N Z_i X_i' \right)^{-1} \frac{1}{N} \sum_{i=1}^N Z_i e_i.$$

Step 2: Apply the WLLN. Because (X_i, Z_i, e_i) are iid with finite second moments:

$$\begin{aligned}\frac{1}{N} \sum_{i=1}^N Z_i X_i' &= E(Z_i X_i') + o_p(1), \\ \frac{1}{N} \sum_{i=1}^N Z_i e_i &= E(Z_i e_i) + o_p(1) = o_p(1),\end{aligned}$$

where the last equality uses the instrument exogeneity condition $E(Z_i e_i) = 0$.

Step 3: Classify the matrix factor. Since $E(Z_i X_i')$ is invertible and $\frac{1}{N} \sum_{i=1}^N Z_i X_i' \xrightarrow{p} E(Z_i X_i')$, the continuous mapping theorem gives

$$\left(\frac{1}{N} \sum_{i=1}^N Z_i X_i' \right)^{-1} = [E(Z_i X_i')]^{-1} + o_p(1) = O_p(1).$$

Step 4: Conclude.

$$\hat{\beta}^{IV} - \beta = O_p(1) \cdot o_p(1) = o_p(1),$$

which establishes $\hat{\beta}^{IV} \xrightarrow{p} \beta$.

5. Asymptotic distribution of $\hat{\beta}^{IV}$.

From the decomposition in Exercise 4:

$$\sqrt{N} (\hat{\beta}^{IV} - \beta) = \underbrace{\left(\frac{1}{N} \sum_{i=1}^N Z_i X_i' \right)^{-1}}_{A_N} \underbrace{\frac{1}{\sqrt{N}} \sum_{i=1}^N Z_i e_i}_{S_N}.$$

Step 1: Convergence of A_N (matrix factor). By the WLLN and CMT (as in Exercise 4):

$$A_N = [E(Z_i X_i')]^{-1} + o_p(1) \xrightarrow{p} [E(Z_i X_i')]^{-1}.$$

In other words, $A_N = [E(Z_i X_i')]^{-1} + o_p(1)$.

Step 2: CLT for S_N (score term). The summands $Z_i e_i$ are iid with

$$\begin{aligned}E(Z_i e_i) &= 0 \quad (\text{instrument exogeneity}), \\ \text{Var}(Z_i e_i) &= E(e_i^2 Z_i Z_i') =: \Omega \quad (\text{finite by assumption}).\end{aligned}$$

The CLT therefore gives

$$S_N = \frac{1}{\sqrt{N}} \sum_{i=1}^N Z_i e_i \xrightarrow{d} N(0, \Omega).$$

Step 3: Combine via Slutsky's theorem. Since $A_N \xrightarrow{p} [E(Z_i X_i')]^{-1}$ (convergence in probability to a constant matrix) and $S_N \xrightarrow{d} N(0, \Omega)$, Slutsky's theorem gives

$$\sqrt{N}(\hat{\beta}^{IV} - \beta) = A_N S_N \xrightarrow{d} [E(Z_i X_i')]^{-1} \cdot N(0, \Omega).$$

For a constant matrix M and $V \sim N(0, \Omega)$, we have $MV \sim N(0, M\Omega M')$, so

$$\sqrt{N}(\hat{\beta}^{IV} - \beta) \xrightarrow{d} N\left(0, [E(Z_i X_i')]^{-1} E(e_i^2 Z_i Z_i') [E(Z_i X_i')]^{-1'}\right).$$

Remark on the asymptotic variance. The expression $[E(Z_i X_i')]^{-1} E(e_i^2 Z_i Z_i') [E(Z_i X_i')]^{-1'}$ is a *sandwich* variance. Under homoskedasticity, $E(e_i^2 Z_i Z_i') = \sigma_e^2 E(Z_i Z_i')$, and the sandwich simplifies. A consistent estimator of the asymptotic variance is obtained by plugging in sample analogues (sample moments and residual-based estimates of σ_e^2).